



DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Assissment 1

Student Name: Sagar Chhetri

Branch: CSE

Semester: 3rd

Subject Name: DAA

UID: 24BCS11059

Section/Group: 707-SAP-B

Date of Performance: 23-01-26

Subject Code: 24CSH-206

1. Aim:

I) Game of Rotation:

Input Format

First line contains N denoting the number of elements on the belt.

Second line contains N space separated integers.

Output Format

Output the required $PMEAN$

Constraints

$1 \leq N \leq 10^6$

$-10^9 \leq \text{each number} \leq 10^9$

For any rotation, $PMEAN$ will always lie within the range of 64-bit signed integer.

2. Requirements (Hardware/Software): HackerRank

3. Procedure/code:

Code:

```
#include <iostream>
#include <vector>
#include <numeric>
#include <algorithm>
```

```
using namespace std;
```

```
int main() {
    int n;
    cin >> n;
    int arr[n];
    int total = 0;

    long long pmeanf = 0;
    for (int i = 0; i < n; ++i) {
        cin >> arr[i];
        total += arr[i];
        pmeanf += (i + 1) * arr[i];
    }

    long long pmeans = pmeanf;
    for (int i = 0; i < n; ++i) {
        pmeanf = pmeanf - total + n * arr[i];
    }

    if (pmeanf > pmeans) {
        pmeans = pmeanf;
    }
    cout << pmeans;

    return 0;
}
```

Screenshot:

```
C++
1 #include <iostream>
2 #include <vector>
3 #include <numeric>
4 #include <algorithm>
5
6 using namespace std;
7 #24bcs11059
8 int main() {
9     int n;
10    cin>>n;
11    int arr[n];
12    int total =0;
13
14    long long pmeanf =0;
15    for(int i=0;i<n;++i) {
16        cin >>arr[i];
17        total +=arr[i];
18        pmeanf+=(i+1)*arr[i];
19    }
20    long long pmeans=pmeanf;
21    for(int i=0;i<n;++i) {
22        pmeanf=pmeanf-total+n*arr[i];
23    }
24    if(pmeanf>pmeans){
25        pmeans=pmeanf;
26    }
27
28    cout <<pmeans;
29
30    return 0;
31 }
32
33
```

Line: 7 Col: 12

4. Output:

Line: 7 Col: 1

☒ Test against custom input

[Upload Code as File](#)

3
20 30 10

Run Code Submit Code

Custom Testcase ⓘ

Compilation Successful
Input (stdin)

3
20 30 10

Your Output

110

5. Learning Outcome:

- Understand array rotation concepts
Learn how circular left rotations work and how data shifts within an array structure.
- Apply problem-solving techniques
Analyze a problem statement, identify constraints, and derive an efficient solution strategy.
- Implement array manipulation in C++
Gain hands-on experience in handling arrays, indexing, and modular arithmetic.
- Optimize logic for performance
Learn to avoid unnecessary rotations by using mathematical observations for efficient computation.
- Work with iterative and conditional structures
Strengthen understanding of loops, condition checking, and control flow in C++.
- Develop algorithmic thinking
Improve ability to reason about sequences, rotations, and optimization problems.
- Handle competitive programming style problems
Gain experience with logic-based challenges similar to those in coding competitions.
- Improve debugging and testing skills
Learn to test edge cases such as large inputs and repeated values.